Optimal Dynamic Pricing With Preferences for Novelty

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November 8, 2026

Motivation

- Firms have more access than ever to consumer information
- Allows for targeted discounting based on preferences derived from repeat interactions
- Despite legislation in the EU, data collected through a purchase is able to collected and stored by the firm (EU, 2026)

How valuable are repeated interactions to the firm?

Research Question

How can firms price discriminate via consumer preferences for novelty and habit?

How do I answer?

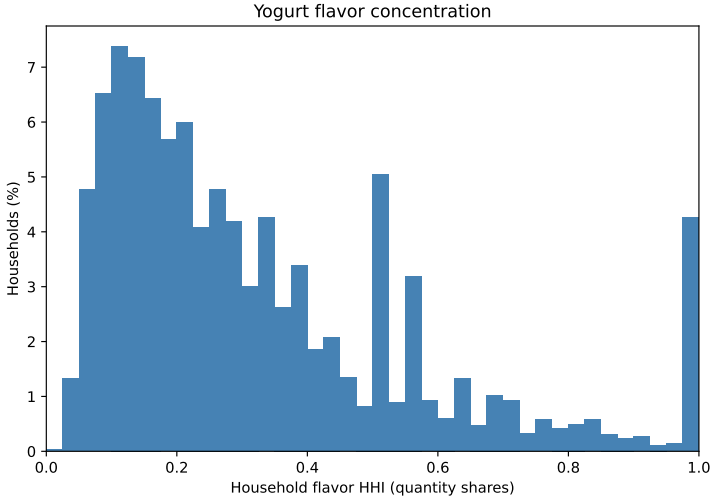
- Solve a theoretical model to understand firm's behavior
- Estimate consumer's preferences with Nielsen data, recovering the distribution of novelty/habit preferences
- Fit that distribution to the supply theory to understand how firm's revenues are impacted by this sort of targeting

Literature

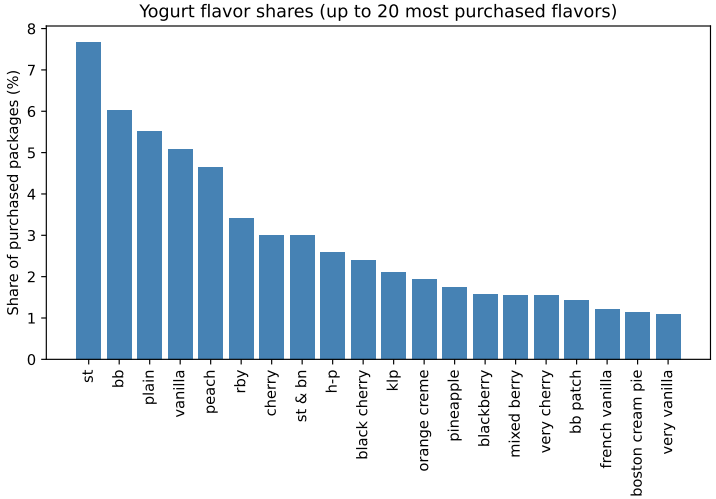
Targeting:

Estimating preferences for novelty:

HHI


Yogurt flavor concentration
Households (%)
Household flavor HHI (quantity shares)
0
1
2
3
4
5
6
7
0.0
0.2
0.4
0.6
0.8
1.0

Flavor Popularity



Supply Setting

- Monopolistic firm producing a single product j
- Firm is profit maximizing, setting prices according to that objective
- A consumer is born into the world with a type $r \in \{H, L\}$, determining how much utility they derive from switching and an initial state $h_0 \in \{0, j\}$
- Consumers live for two periods, purchasing the inside good or opting for the outside good in each period
- The firm knows initial states and types, setting prices to individual consumers by type and history

Consumer's Problem

$$V_{jt}^{inside} = \underbrace{\beta}_{\text{Constant}} + \overbrace{\gamma_r \mathbb{1}\{h=0\}}^{\text{Utilty from Switching}} - \underbrace{\alpha p_t(h)}_{\text{Price Disutility}}, \quad V_{0t}^{Outside} = \underbrace{\gamma_r \mathbb{1}\{h=j\}}_{\text{Utility from Switching}}$$

$$v_t(x) = V_{jt}^{inside} - V_{0t}^{outside} = \beta + \underbrace{\gamma[\mathbb{1}\{h=0\} - \mathbb{1}\{h=j\}]}_{q(h)} - \alpha p_t(h)$$

- Prices, set by the firm, and switching utility depend on history $h = x_{t-1}$
- Utility for the outside good is normalized to zero

Firm's Problem

$$\pi_t(h) = \max_{p_t(h)} s_t(h)[p_t(h) - c]$$

$$\Pi(h) = \underbrace{s_1(h)(p_1(h) - c)}_{\text{Margin in Period 1}} + \underbrace{\delta\{s_1(h)\pi_2(j) + (1 - s_1(h))\pi_2(0)\}}_{\text{Continuation Value}}$$

- Firm profits are comprised of period one margins, given history of consumption, and the value of that consumer in the different states in period two
- The firm will set prices according to consumer type and history to maximize expected profit

Optimal Terminal Price and Profit

$$\frac{\partial \pi_2(h_1)}{\partial p_2(h_1)} = s_2(h_1) + (p_2 - c)\frac{\partial s_2(h_1)}{\partial p_2(h_1)}$$

$$p_2^*(h_1) = c + \frac{1}{\alpha(1 - s_2^*(h_1))}$$

$$\pi_2^*(p_2^*(h_1); h_1) = \frac{s_2^*(h_1)}{\alpha(1 - s_2^*(h_1))}$$

- Optimal period two price is cost plus a demand markup term
- Period two profit at optimal prices is the odds of purchase, scaled by the inverse price sensitivity

First Period Optimization

$$\frac{d\Pi}{dp_1(h_0)} = s_1 + \frac{\partial}{\partial p_1(h)} s_1(h_0)[p_1(h_0) - c + \delta(\pi_2(j) - \pi_2(0))]$$

$$p_1^*(h_0) = c + \frac{1}{\alpha(1 - s_1^*(h_0))} - \delta(\pi_2(j) - \pi_2(0))$$

$$\Pi^*(h) = \delta\pi_2(0) + \frac{s_1^*(h_0)}{\alpha(1 - s_1^*(h_0))}$$

Optimal Profit Given Switching Type

$$\frac{d\Pi}{d\gamma} = \frac{q(h)s_1^*(h) - \delta s_1^*(h)s_2^*(j) + \delta(1 - s_1^*(h))s_2^*(0)}{\alpha}$$

- There are two interpretations:
  1. If $h_0 = j$, $q(h) < 0$, leading us to ambiguous signage as we raise profits from initially outside consumers but see opposing forces for the initially inside: leave in period one, return in period two
  2. If $h_0 = 0$, $q(h) > 0 \Rightarrow \frac{d\Pi}{d\gamma} > 0$, meaning that profits increase as γ increases
- Points to firms potentially reaping great benefit from knowing consumer types and histories

Pricing Simulations Specification

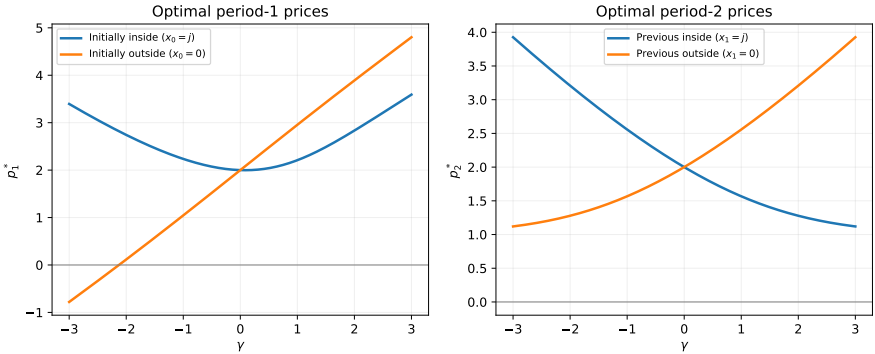
Parameter	Description	Value
β	Inside-good intercept	2
α	Price sensitivity	1
δ	Discount factor	0.92
c	Marginal cost	0
γ	Switching preference	$[-3, 3]$
Initial Consumption State		$h_0 \in \{0, j\}$
Feasible Prices		$p_t(h) \in \mathbb{R}$

- Allow for negative pricing
- Firm has knowledge of both type and history, allowing for targeted pricing
- Four consumption histories possible $(h_0, h_1) \in \{(0,0), (0,j), (j,j), (j,0)\}$

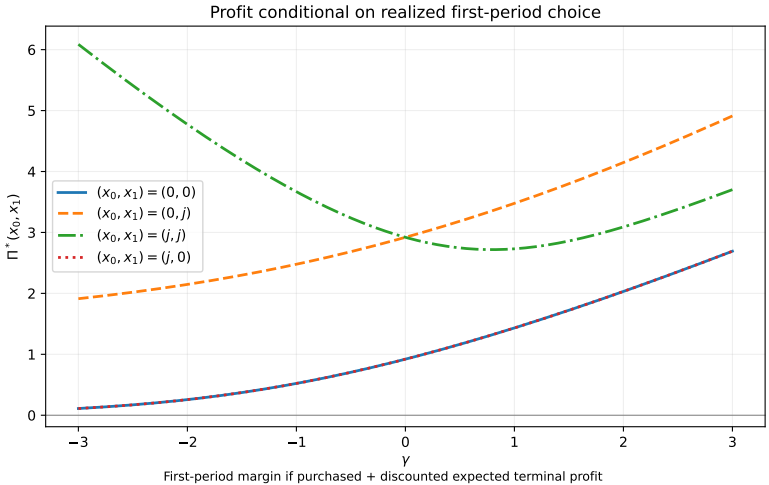
Results

x_0	γ	p_1^*	$p_2^*(j)$	$p_2^*(0)$	Π^*	$\Pi^{*\prime}$
j	-3	3.395	3.926	1.120	5.087	-1.387
j	-1	2.236	2.557	1.567	2.669	-0.959
j	0	2.000	2.000	2.000	1.920	-0.500
j	1	2.209	1.567	2.557	1.731	0.125
j	3	3.592	1.120	3.926	2.702	0.668
0	-3	-0.780	3.926	1.120	0.913	0.195
0	-1	1.045	2.557	1.567	1.478	0.385
0	0	2.000	2.000	2.000	1.920	0.500
0	1	2.956	1.567	2.557	2.478	0.615
0	3	4.801	1.120	3.926	3.912	0.804

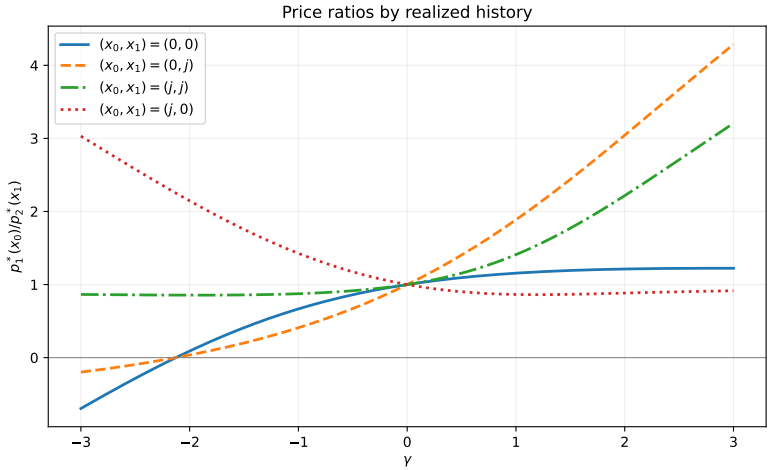
Optimal Prices



Optimal Profit



Optimal Price Ratio



Discussion

- Two mechanisms: Induce and harvest, Retain and gain
- Most profit from those who are habitual inside purchasers
- Least profit from those who purchase outside in period 1 across all levels of gamma
- Interesting when (j, j) and $(0, 0)$ cross

Data Sources

Nielsen HomeScan

- Provides household level purchase data
- Covers years 2006-2015
- Products chosen, quantity chosen, timing of trips

Nielsen RetailScan

- Provides store level product data
- For specified product category: prices, markets, product attributes

Sample Description

- Must be single-person household
- Must be seen at least 2 times
- Purchases must be made at Nielsen store (prices available)
- Flavor attribute must be present

Sample Statistics: Purchase Activity

	Full sample	Yogurt sample
Households	37,827	15,106
Observed panel years	3.96	3.16
Trips per household	493.13	20.26
Trips per observed week (approx.)	2.27	0.11
Distinct UPCs per household	1,178.31	13.89
Purchase lines per household	3,397.64	45.15
Packages per household	4,669.07	93.77

All statistics except household counts are means.

Sample Statistics: Demographics

	Full sample	Yogurt sample
Head age (category code)	6.96	7.01
Household income (category code)	16.18	17.06
White (%)	80.33	81.58
Black (%)	12.81	12.15
Asian (%)	1.57	1.57
Other race (%)	2.19	2.11
Hispanic (%)	3.10	2.59

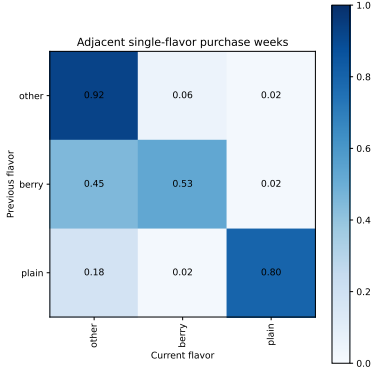
Age and income are mean category codes.

Switching Stats

Table: Matched yogurt sample: purchase dynamics

metric	value
Matched households	15106.00
Observed yogurt purchase weeks	282648.00
Mean flavors bought per household	10.76
Mean household flavor HHI (0 to 1)	0.32
Mixed-flavor weeks (%)	37.88
Weeks above seven packages (%)	21.72
Eligible adjacent single-flavor week pairs	30392.00
Switch rate in eligible pairs (%)	46.75
Mean single-flavor spell length (consecutive calendar weeks)	1.10
Median gap between purchase weeks	3.00

Persistence Across Aggregated Flavor Groups



Plain repeats in approximately 80% of eligible transitions, versus 53% for berry.

Demand Setting and Objective

- Consumers can either purchase the inside good, a flavor of yogurt, or the outside good, not yogurt
- Following Nevo (2001), a week contains seven choice opportunities with observed flavor purchases treated as fulfilling an opportunity
- Define outside good as $\max(0, 7 - \sum_j x_{jt})$, or the difference in purchases and a full week
- Consumers differ in preferences for novelty and price sensitivity, but derive the same base utility from the three flavor categories
- Goal: Recover distribution of novelty preferences

Demand Model

$$U_{ijt} = \overbrace{\beta_0}^{Constant} + \underbrace{\beta_j X_{jt}}_{\text{Flavor specific constant}} + \overbrace{\gamma_i N_{ijt} + \phi_i H_{ijt}}^{\text{Preferences for novelty/habit}} + \underbrace{\alpha_i p_{jt}}_{\text{Price disutility}} + \overbrace{\varepsilon_{ijt}}^{\text{T1EV}}$$

$$\text{s.t. } N_{ijt} = \mathbb{1}\{x_{t-1} \neq j\},\ H_{ijt} = \frac{\sum_{t=1}^{t-1} \mathbb{1}\{x_{it-1} = j\}}{\sum_{t=1}^{t-1} \mathbb{1}\{x_{it-1} \neq 0\}}$$

- N_{ijt} represents a novel choice in the prior period, buying a flavor other than j
- H_{ijt} represents repeated purchase, the share of total purchases which flavor j comprises
- Normalize outside option to zero s.t. $U_{i0t} = \varepsilon_{i0t}$

Probability and Shares

$$P_{ijt} = \frac{\exp\{V_{ijt}\}}{1 + \sum_{k \in J} \exp\{V_{ikt}\}}$$

$$\text{s.t. } V_{ijt} = U_{ijt} - \varepsilon_{ijt}$$

$$s_{jt} = \frac{1}{N} \sum_{n=1}^{N} P_{ijt}$$

- Use deterministic utility to form purchase probabilities
- Take average probability for product j across individuals $i \in N$ to recover shares

Estimation

$$P_i(\theta) = \int P(y_i \mid \nu, \theta)\ \phi(\nu)\, d\nu$$

$$\widehat{P}_i(\theta) = \frac{1}{64} \sum_{r=1}^{64} \prod_{t=1}^{T_i} P_(y_i \mid \nu_{ir}, \theta) \qquad \nu_{ir} \sim \mathcal{N}(0, I)$$

$$Q_{64}(\theta) = -\sum_{i=1}^{N} \log \widehat{P}_i(\theta)$$

- ν_i captures unobserved heterogreneity, y_{it} represents observed choice, $\theta = \{\beta, \gamma, \alpha\}$ is the vector of heterogeneous demand parameters
- 64 draws per household from a normal distribution
- 3 starts at different starting values, keep lowest objective
- Minimize negative log-likelihood

Estimates

Table: Demand estimates

Parameter	Estimate	Standard error
Inside constant b_0	1.1969	0.0034
Berry vs. other β_{berry}	-0.6143	0.0034
Plain vs. other β_{plain}	-0.2689	0.0055
Mean price coefficient $E[\alpha_i]$	-0.9206	0.0051
Mean novelty coefficient γ_i	-0.8968	0.0096
Mean habit effect ϕ_i	0.0518	0.0003
SD price coefficient $\text{SD}(\alpha_i)$	0.9781	0.0117
SD novelty coefficient	0.8358	0.0093
SD own-flavor stock coefficient	0.0751	0.0004

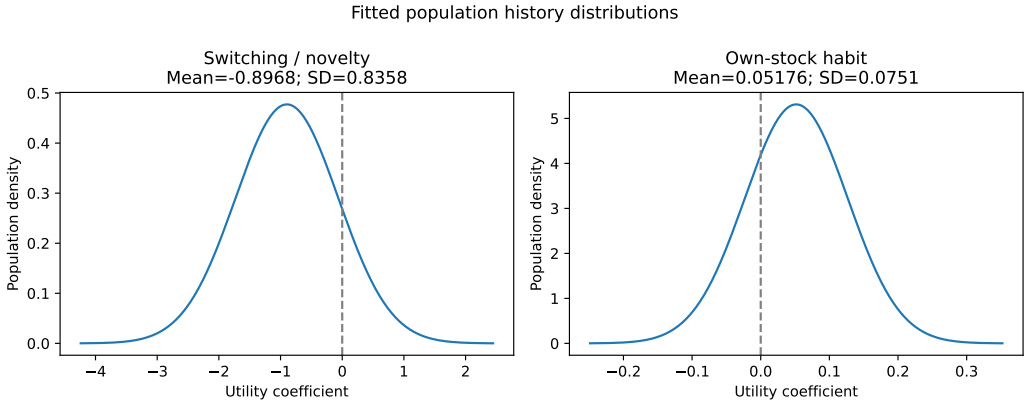
Implied Willingness to Pay

Table: Implied willingness to pay

Utility coefficient	Mean WTP	Population SD
Inside constant b_0	2.7681	2.9412
Berry vs. other β_{berry}	-1.4208	1.5096
Plain vs. other β_{plain}	-0.6220	0.6609
Novelty	-2.0741	3.5793
Own-flavor stock	0.1197	0.2836

- Would need to be subsidized to switch, though lots of variation
- Willing to pay to continue consuming their same product

Distributions



Discussion

- Mean novelty effect is equivalent to $\approx \$2$ increase
- Habit provides some utility
- Both plain and berry are less preferred than other flavors

How can the firm use the distribution of switching types to price discriminate?

Connecting Supply and Demand

Supply Theory

- Today's pricing offer changes tomorrow's demand
- Single product firm pricing to attract customers against outside option
- Maintain two-period timing with initial state

Empirical Demand

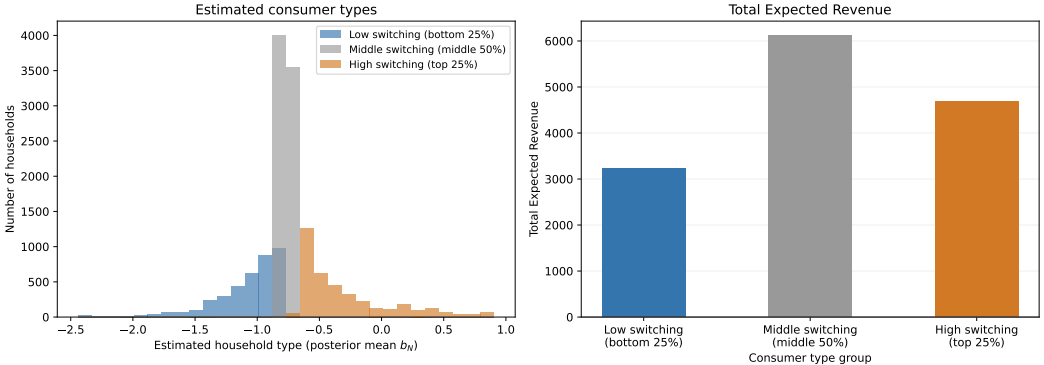
- Employ estimated preference for novelty
- Access observed purchase histories
- Novelty and habit both enter demand

How well does the theory map to what is observed in the data?

Procedure

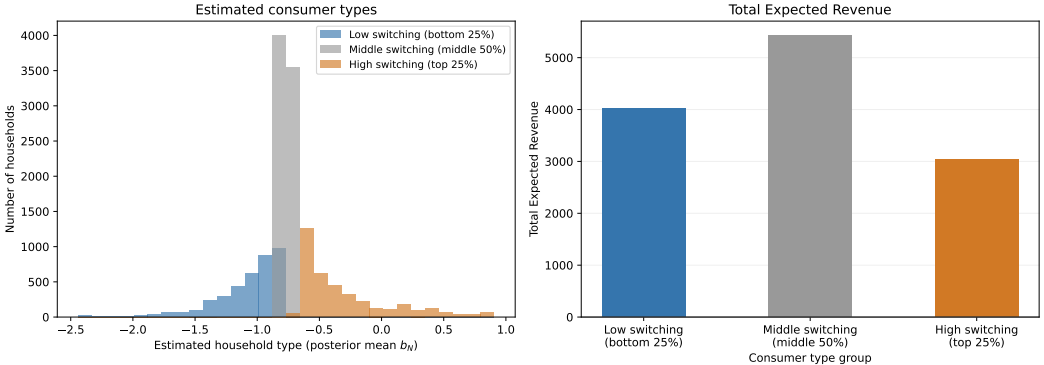
- Recover household type from observed purchase history and treat estimated type as known to firm
- Use initial purchase y_{i1} to determine "initial state", h_0
- Separate high and low types as the 75th and 25th percentiles, respectively
- Use observed prices and simulate discounts based on them, specific to the different types and their consumption path
- Focus on berry and plain to maintain parity with the theoretical model and avoid aggregate weighting
- Use estimated type γ_i to discriminate, choosing an initial price offer, observing the period one choice, and choosing the terminal price

Results: Berry Inside



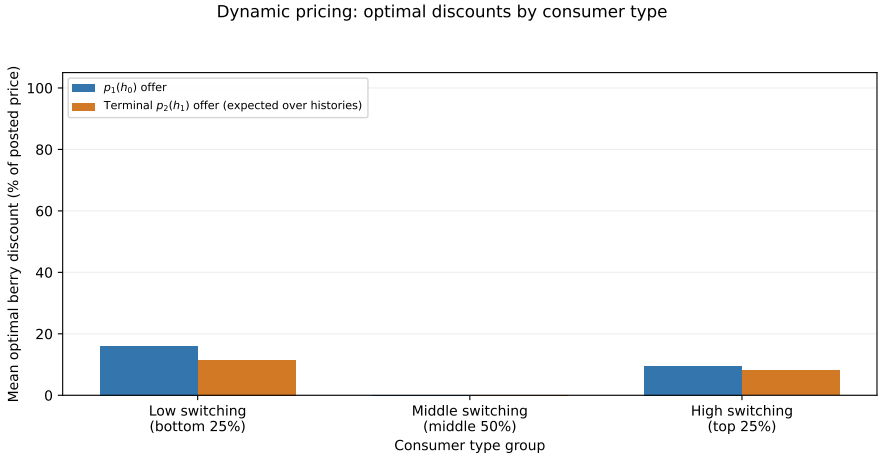
First-period revenue + 0.92 × expected terminal revenue. Low/high types receive personalized berry offers; middle types receive no coupons.

Results: Plain Inside



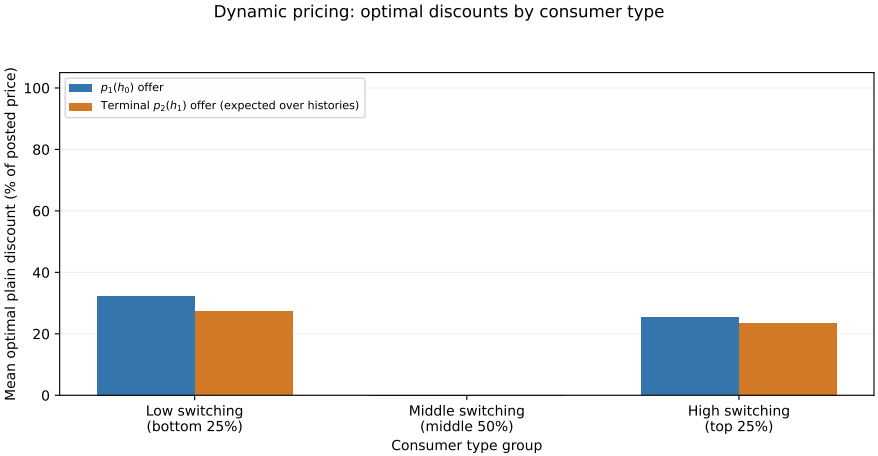
First-period revenue + 0.92 × expected terminal revenue. Low/high types receive personalized plain offers; middle types receive no coupons.

Berry: Optimal Offers by Consumer Type



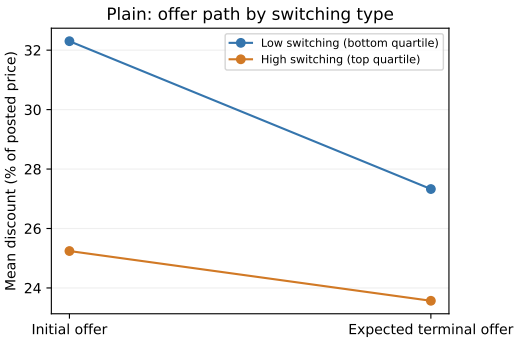
Household-weighted means among berry-eligible households. Middle group receives no coupons by policy design.

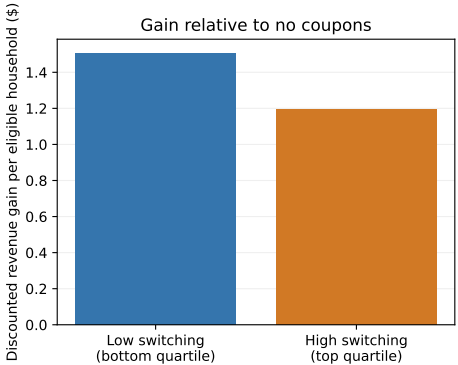
Plain: Optimal Offers by Consumer Type



Household-weighted means among plain-eligible households. Middle group receives no coupons by policy design.

Plain: Initial Offers and Subsequent Pricing





Terminal offers average over purchase histories. Type groups also differ in price sensitivity and habit.

Comparison

Table: Focal flavor comparison

	Plain		Berry	
	No coupon	Personalized	No coupon	Personalized
Period 1 revenue	5,914.97	6,518.23	6,880.61	7,228.55
Period 2 revenue	5,714.51	6,492.98	6,921.30	7,414.50
Discounted revenue	11,172.32	12,491.77	13,248.21	14,049.90
Baseline revenue	11,172.32	11,172.32	13,248.21	13,248.21
Revenue change	0.00	1,319.45	0.00	801.69
Revenue change (%)	0.00	11.81	0.00	6.05
Period 1 focal units	3,104.96	3,947.93	7,524.28	8,179.01
Period 2 focal units	3,077.82	3,906.07	7,664.87	8,293.98
Coupon expenditure	0.00	3,640.83	0.00	2,461.53

Conclusion

- Firm can gain up to 12% simulated revenue by targeting based on switching type relative to no coupons
- Consumers experience disutility from switching, needing to be subsidized $\approx \$2$ to switch products
- Consumers enjoy habit and would be willing to pay $\$0.12$ to contribute another unit to the product's share of purchase history

Next Steps

- Add firm learning to supply side
- Extend market structure on supply
- Add more product attributes to empirical estimation
- Estimate unobserved quality in BLP fashion

Shares and Elasticities

$$s_t(h) = \frac{e^{v_t(h)}}{1 + e^{v_t(h)}}, \; s_{0t}(h) = 1 - s_t(h)$$

$$\frac{\partial s_t(h)}{\partial p_t(h)} = -\alpha s_t(h)(1 - s_t(h))$$

$$\frac{\partial s_t(h)}{\partial \gamma} = q(h) s_t(h)(1 - s_t(h))$$